Healthcare Analytics ASC Workshop · Handout

*Healthcare analytics uses the huge amount of data in healthcare to help achieve the **Quintuple Aim**. Circle the terms you hear today; star the ones you want to learn more about.*

The Big Picture

Quintuple Aim Five goals for health care: patient experience, population health, lower costs, care team well-being, and health equity.
Payer / Payor The organization that pays for care (an insurer, Medicare, Medicaid).
Provider Whoever delivers care: a doctor, clinic, or hospital.
Payor-provider (integrated system) One organization that both insures and delivers care (Highmark + AHN, Intermountain + Select Health).
Value-based care Paying providers for good outcomes instead of for each service.

Data Sources

Claims Bills a provider sends to a payer: who, what, when, where, and how much.
EHR / EMR Electronic Health (Medical) Record: the clinical chart with labs, vitals, notes.
ACS American Community Survey: the Census Bureau's yearly survey of neighborhoods.
SVI Social Vulnerability Index (CDC): 0–1 score of how vulnerable an area is.
SDOH Social Determinants of Health: food, housing, transportation, income, etc.
Vendor data Data bought from a company (e.g., household-level SDOH or consumer data).

IDs & Linking

Member ID The insurer's ID for a person.
MRN Medical Record Number: the health system's ID for a patient.
Crosswalk A table that maps one ID to another (MRN ↔ member ID).
Match rate Share of people a vendor could link to its data.
Grain What one row represents (a claim line, an encounter, a ZIP).

Codes

ICD-10 International Classification of Diseases: diagnosis codes (E11.9 = type 2 diabetes).
CPT / HCPCS Procedure codes: what was done (99213 = office visit).
NDC National Drug Code: identifies a specific drug.
POS Place of Service code (11 office, 23 ED, 21 inpatient, 02 telehealth).
DRG Diagnosis-Related Group: how hospital stays are grouped and paid.

Money & Utilization

Allowed / Paid amount What the plan agreed the service is worth / what it actually paid.
Reversal A canceled claim, recorded as negative dollars; net these out first.
Member-months Number of months each person was enrolled, added up.
PMPM Per Member Per Month: total cost ÷ member-months.
TCC Total Cost of Care: all medical + pharmacy spending for a person.
ED / IP / LOS Emergency department · inpatient stay · length of stay.
Readmission Returning to the hospital soon (often within 30 days) after discharge.

Plans & Quality

Commercial Insurance through an employer or bought privately.
MA Medicare Advantage: private plans for people 65+ (or with disabilities).
Medicaid Public insurance for people with low income.
HEDIS / Stars Standard quality measures used to rate health plans.
Risk adjustment Accounting for how sick members are when comparing cost or quality.

Clinical Measures

BMI Body Mass Index: weight (kg) ÷ height (m)².
A1c Average blood sugar over ~3 months; ≥ 6.5% suggests diabetes.
PHQ-9 9-question depression screener, scored 0–27.
GLP-1 RA A class of diabetes/weight-loss drugs (e.g., semaglutide).

Methods

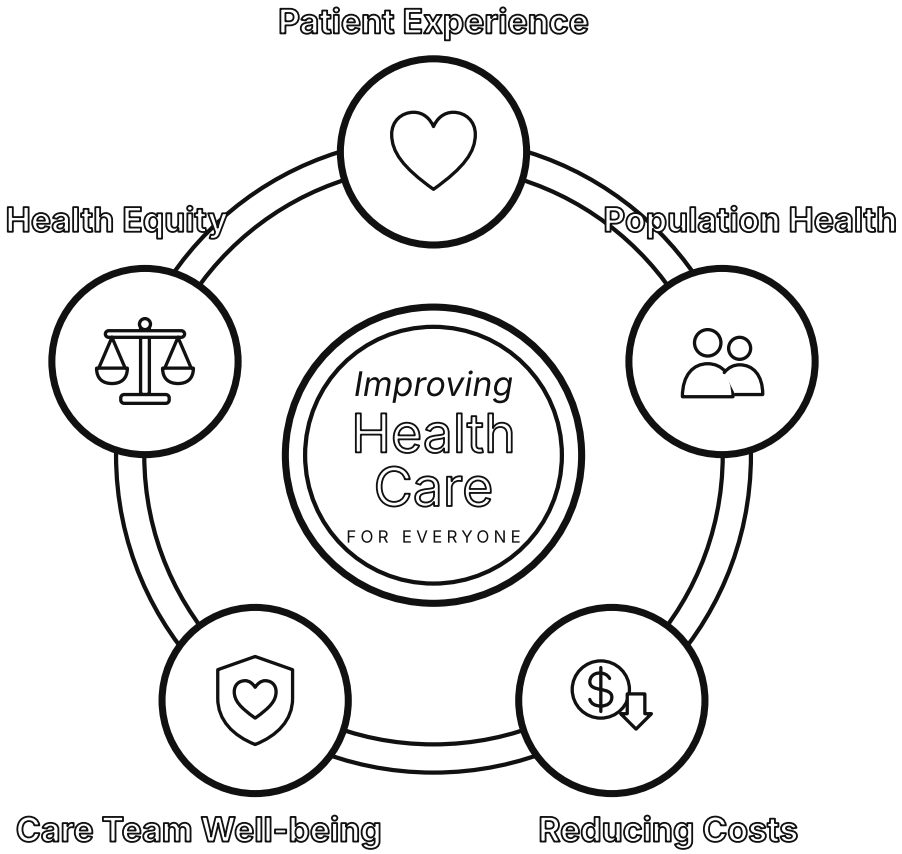
Ecological fallacy Assuming what's true of an area is true of each person in it.
Poisson regression A model for counts (like ED visits); use an **offset** for exposure time.
IRR Incidence Rate Ratio: how many times higher the rate is (1 = no difference).
Propensity score Probability of getting a treatment; used to compare similar groups.
DiD Difference-in-Differences: compares change over time vs a comparison group.

Claims & EHR

Census & Vendor

Color the Quintuple Aim

Pick one color per aim, then circle the aim you would most like to work on.



The Four Data Sources

Color each icon, then write one question that source could answer.

Claims
bills sent to the insurer



EHR
the patient's clinical chart



Census / Public
neighborhoods & communities



Vendor
purchased data on people



One thing I learned

A question I still have

A job I'd want